Data 37711: Machine Learning & AI Part I **Fall 2026**

Homework 3

Lecturer: Victor Veitch Due date: TBD

1. Suppose that Y is in the natural numbers. You decide to model Y given X with boosting. As your base model, you decide to use regression trees $\{h_j\}$ that take in covariates X and output real valued entries. Then,

   (a) You decide to make your model $P(Y|X) \sim \text{Poi}\left(\exp\left(f(X)\right)\right)$, where $f(X) = \sum_j \lambda_j h_j(X)$, with each $h_j(X)$ a (trained) regression tree.

      i. Why is Poisson a reasonable choice here?

      ii. Why do we use exp() as our link function?

   (b) Write down the loss function used for learning in terms of $f(x)$ and Y.

   (c) Give an expression for $\frac{dL}{df(x)}$.

   (d) Describe how you would choose each h_j in (first order) boosting.

2. Often, machine learning models involve so-called *hyperparameters*. These are quantities that are set by the analyst, rather than being learned by empirical risk minimization. For example, the regularization strength is a hyperparameter. One option for selecting hyperparameters is to try fitting a model with several different choices, and then using performance on a test set to select the best one.

   (a) Find (and prove) a non-trivial generalization bound for this procedure. That is, find a probabilistic bound on the gap between the risk estimated on the test set, and the true risk of the model. Give this bound in terms of (some subset of) the number of training examples n, the number of examples in the test set l, and the number of hyperparameters k.

   (b) What might go wrong if you use the same test set to select hyperparameters and to estimate the generalization error? In particular, consider the case where k is large relative to l.

   (c) Commonly, we instead split the data into a training set, a test set, and a *validation* set used to select hyperparameters.

   Use the provided dataset on white wine (`winequality-white.csv`, a subset of the wine quality dataset) and divide it randomly into 70% train, 10% validation, and 20% test. The last column "quality" is the target, an integer from 1–10. We'll view this as a regression problem with mean squared error as the desired loss function. Pick two model classes from the set { RandomForestRegressor, KNeighborsRegressor, MLPRegressor } and read their documentation in Scikit-learn to understand the hyperparameters of these models and what they do. Do grid search over a reasonably-large set of parameters (number of parameter sets > the number of points in the validation set): fit the model on the train set for each set of parameters, and select the "best parameters" using the score on the validation set. Finally, report the performance on the test set for each of the model classes you chose. What's the relationship between the empirical risk on the validation set and the empirical risk on the test set?

3. In class, we talked about the generalization behaviour of boosting as an increasing number of weak classifiers are included in the model fitting.

   (a) Use scikit-learn to fit a gradient boosting model to dset0 (You should model the first column in `dset0.csv` given the other columns). Fit models with 1, 10, 100, 300, and 1000 trees. What do you observe about the generalization behaviour of these models?

   (b) Use scikit-learn to fit a gradient boosting model to dset1 (You should model the first column in `dset1.csv` given the other columns). Fit models with 1, 10, 100, 300, and 1000 trees. What do you observe about the generalization behaviour of these models?

   (c) If your observations in parts (a) and (b) differ, explain what property of the data might explain that.